Title

Multiple Sequence Alignment of Hemoglobin Beta Proteins from Five Mammalian Species Using Clustal Omega

Objective

The objective of this task was to perform a multiple sequence alignment (MSA) of five hemoglobin beta protein sequences from different mammalian species using Clustal Omega. The analysis aimed to identify conserved amino acid residues, sequence similarities, and evolutionary relationships among the selected proteins.

Introduction

Hemoglobin beta (HBB) is an essential protein responsible for oxygen transport in vertebrates. Due to its crucial biological function, the beta-globin protein is highly conserved across mammalian species. Multiple Sequence Alignment (MSA) is commonly used in bioinformatics to compare homologous protein sequences, identify conserved regions, and infer evolutionary relationships.

Materials and Methods

Five hemoglobin beta protein sequences were obtained from the UniProt database:

- *Homo sapiens* (Human)
- *Mus musculus* (Mouse)
- *Bos taurus* (Cow)
- *Sus scrofa* (Pig)
- *Canis lupus familiaris* (Dog)

The sequences were downloaded in FASTA format and combined into a single FASTA file. Multiple Sequence Alignment was performed using the Clustal Omega online server with default parameters.

Results

The alignment demonstrated a high degree of sequence conservation among all five hemoglobin beta proteins.

Most amino acid positions were identical, indicating that the protein has been strongly conserved during mammalian evolution. Conserved residues were represented by asterisks (*) in the alignment, whereas conservative amino acid substitutions were indicated by colons (:) and periods (.).

The guide tree and phylogenetic tree grouped the proteins according to their sequence similarity. Human and dog hemoglobin beta proteins clustered closely together, while pig and cow proteins showed moderate similarity. Mouse hemoglobin beta displayed the greatest evolutionary distance among the analyzed sequences.

Discussion

The high level of sequence conservation reflects the essential biological role of hemoglobin beta in oxygen transport. Only a limited number of amino acid substitutions were observed, suggesting that mutations within this protein are constrained by functional requirements.

The phylogenetic analysis supports the expected evolutionary relationships among mammals and demonstrates that homologous proteins retain significant similarity despite species divergence.

These findings illustrate the usefulness of Multiple Sequence Alignment for identifying conserved protein motifs and studying evolutionary relationships.

Conclusion

The multiple sequence alignment successfully identified highly conserved regions within hemoglobin beta proteins from five mammalian species.

The analysis confirmed that hemoglobin beta is an evolutionarily conserved protein, and the generated guide tree and phylogenetic tree reflected the close evolutionary relationships among the selected mammals.

Overall, Clustal Omega proved to be an effective tool for comparative protein sequence analysis.

Figure Captions

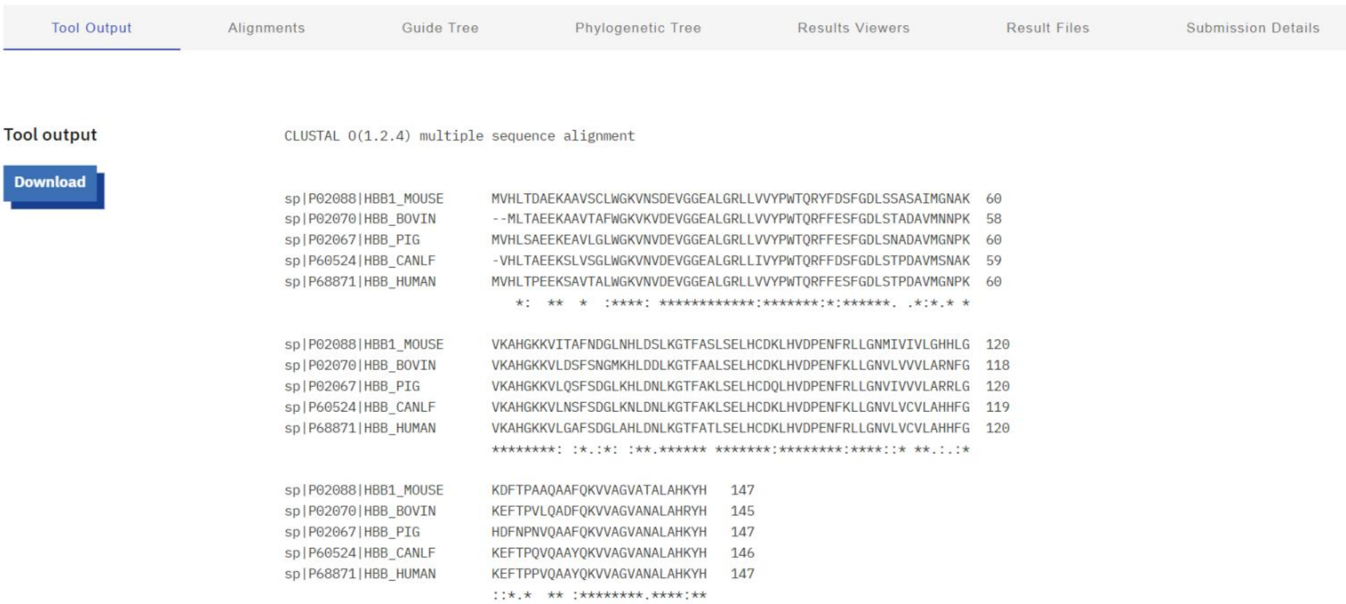

Figure 1. Multiple sequence alignment of hemoglobin beta proteins from five mammalian species generated using Clustal Omega. Conserved amino acid residues are indicated by asterisks (*), while conservative substitutions are marked by colons (:) and periods (.).

Figure 2. Guide tree illustrating the similarity relationships among the five aligned hemoglobin beta protein sequences.

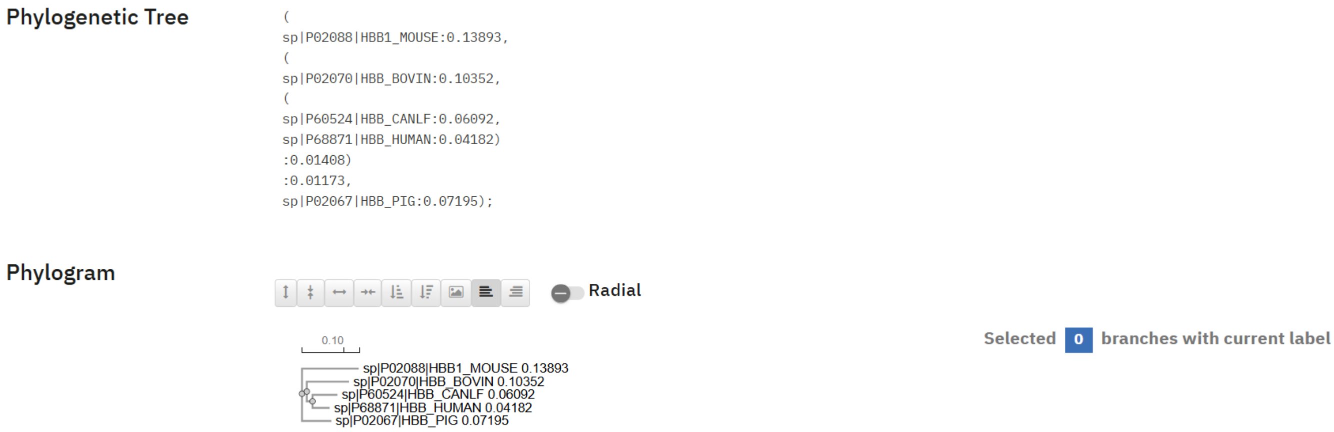

Figure 3. Phylogenetic tree generated from the multiple sequence alignment, showing the evolutionary relationships among the selected mammalian species.